

b-secure Backend — Documentation Index

Backend: Django 5.x + Django REST Framework + Django Channels
Language: Python 3.12+ **Status:** v1.0.0

Documents in This Folder

File	What It Covers
<u>overview.md</u>	Project structure, local setup, settings architecture, URL routing, code conventions
<u>authentication.md</u>	OTP login, JWT tokens, Google/Apple social auth, WebSocket auth middleware
<u>models.md</u>	All Django models across every app, FSM states, DB indexes, migration strategy
<u>api_endpoints.md</u>	All REST API endpoints with request/response JSON examples, ViewSet code
<u>realtime.md</u>	Django Channels WebSocket consumers, Redis channel layer, location tracking protocol
<u>payments.md</u>	Wallet system, Razorpay/Stripe top-up, booking payments, payouts, invoice PDF
<u>notifications.md</u>	FCM push, SMS (Twilio), email (SendGrid), in-app notifications, all task types
<u>sos_and_safety.md</u>	SOS trigger flow, escalation logic, check-in system, dead man's switch, incidents
<u>celery_tasks.md</u>	All Celery tasks, queue architecture, beat schedule, reliability patterns

File	What It Covers
deployment.md	Docker, Docker Compose, AWS infrastructure, Nginx, GitHub Actions CI/CD
testing.md	pytest setup, model/API/service/WebSocket tests, factories, coverage

Quick Reference

Start Development

1. Start infrastructure

```
docker-compose up -d db redis
```

2. Install dependencies

```
python -m venv venv && source venv/bin/activate
pip install -r requirements/development.txt
```

3. Run migrations

```
python manage.py migrate
```

4. Start API server

```
daphne -b 0.0.0.0 -p 8000 config.asgi:application
```

5. Start Celery worker (separate terminal)

```
celery -A celery_app worker -Q high_priority,default -l info
```

6. Start Celery beat (separate terminal)

```
celery -A celery_app beat -l info --scheduler
    django_celery_beat.schedulers:DatabaseScheduler
```

Key API Endpoints

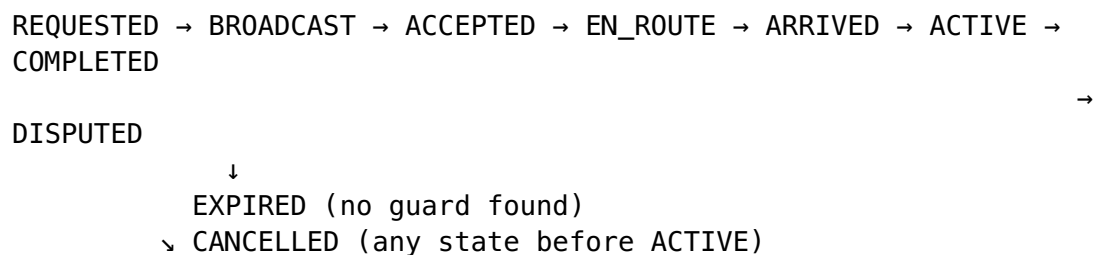
Purpose	Method	Endpoint
Send OTP	POST	/api/auth/send-otp/
Verify OTP	POST	/api/auth/verify-otp/
Create booking	POST	/api/bookings/
Trigger SOS	POST	/api/sos/trigger/
Wallet top-up	POST	/api/payments/wallet/topup/initiate/
Live tracking	WSS	/ws/tracking/{booking_id}/?token=...
Admin SOS feed	WSS	/ws/sos/feed/?token=...

Purpose	Method	Endpoint
Admin dashboard	WSS	/ws/admin/dashboard/?token=...

Django Apps

apps/	
— authentication	OTP, JWT, social auth
— users	UserProfile, Address, EmergencyContact
— guards	GuardProfile, GuardDocument, availability
— bookings	Booking FSM, session OTP, broadcast
— tracking	LocationSnapshot, WS consumers
— payments	Wallet, transactions, payouts, invoices
— notifications	FCM, SMS, email, in-app
— sos	SOSAlert, Incident, escalation
— reviews	Post-session ratings
— admin_panel	Admin-only APIs and WS consumers
— analytics	Aggregated daily stats

Booking States



Run Tests

```
pytest --cov=apps --cov-report=term-missing -v
```

Related Documentation

- [System Requirements](#) — Full product requirements document